

Jackie Gao

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Cambridge MA / Calgary AB

Portfolio Link: https://drive.google.com/file/d/1krtsZ3GyCqLNIDCIU9L4x6QyWiJ5ToeT/view?usp=share_link

Highlights

Cross-disciplinary system problem solving: mechanical, fluidics, thermal, optics, biochemical, data, manufacturing.
Solution delivery in prototyping and product development.
Product reliability in medical-device and SLA 3D-printing.
Project leadership in early stage multidisciplinary teams.

Education

University of Alberta *Edmonton, AB*
B.Sc. in Mechanical Engineering [co-op] 2013-2018
GPA Average: 3.5/4.0 | GPA 4th Year: 3.8/4.0

Competencies

3D Design: SolidWorks, OnShape, AutoCAD, Autodesk Inventor
Technology: AI (Terminal, API), Git, Python, HTML, SQL
Data: Data Analysis, Visualization, Interpretation, Knowledge Bases.
Simulation: Solid Simulation (FEA), Fluidic Simulation (CFD)
Languages: English (Native), Mandarin (Advanced)

Study Abroad

National University of Singapore – *Singapore - Summer 2019*
University of Texas at San Antonio – *Texas, USA - Summer 2018*
RWTH Aachen University – *Aachen, Germany - Summer 2016*

Experience

AI-Enabled Engineering, Prototyping, and Data Analysis

Calgary / Boston (Remote)

Self Directed Research & Development

May 2025 – Present

- Built AI-assisted workflows to quickly iterate from an idea, to research, conception, data verification, concept design, and software prototyping, with AI enhanced documentation and quick information retrieval from messy historical documentations.
- Built and validated an AI automated job application system but decided not to deploy it as of today.
- Prototyped a Google-Maps powered Lexington walking trails app, with a custom software tool to import trail data from PDF maps.
- Conducted AI-powered data exploration in economics and land-use finding strong correlations between wealth/land-ownership.

Microfluidics Systems Integration Engineer + Operational Improvements

Mississauga, ON

Vital Biosciences Inc.

May 2023 - May 2025

- Supported root-cause investigation and optimization of a microfluidics-based blood-testing medical device, improving assay performance through investigations across fluidics, thermals, optics, biochemistry, electrical, data pipelines, and manufacturing.
- Characterized thermal gradients across alpha and beta instrument variations, experimentally verifying air-heater vs. radiative-heater design differences as a key contributor to on-disc assay performance variations thus informing future design directions.
- Root-caused readout data anomalies as manufacturing-induced blemishes in microfluidic readout optical windows.
- Built visualization experiments using dyed fluids to study mixing behavior and cell debris transport inside the microfluidic device.
- Supported system update validation, maintenance, software-debugging, and workflow changes for microfluidics test systems.
- Led integration of precision SLA printing capability to engineering operations, improving engineering prototyping capacity.
- Optimized a PCB stroboscope with electrical and photonics engineers to improve user visualization of spinning microfluidic discs.
- Visited an injection molding supplier in Shenzhen to investigate a reliability issue improving chip reliability from ~80% to ~90%.

Lab Automation Engineering Contractor - Assay Optimization

August 2021 – September 2021

- Optimized and automated an assay on the TECAN liquid handler improving precision from 60% to 90% CV.

Medical Device R&D Engineer

March 2020 – September 2020

- Developed a prototype electromechanical-optical instrument for a centrifugal microfluidic lab-on-chip blood-testing device, integrating motor control, laser-synchronized stroboscopic imaging, spectroscopic readout, and LabJack data acquisition.
- Designed custom sonication and magnetic pulldown lab tools specified for in-house assay development requirements.
- Co-developed a self-operated mechanism to collect blood from the vein, and built a mechanical antecubital vein replica for testing.
- Prototyped magnetic bead separators for a robotic lab automation platform, improving in-house assay bead recovery by 150%.

Sustaining Engineer - Stereolithography 3D Printing Products

Somerville, Massachusetts

Formlabs

October 2021 – November 2022

- Characterized the largest Form 3L tank failure mode, translating findings into design guidelines for improved tank reliability.
- Resolved a chemical incompatibility failure-mode due to manufacturing QC, lowering dead-on-arrival rates from 50% to 0%.
- Conducted lifetime testing of cost-down components, supporting product BOM reduction efforts.
- Coordinated extensive beta testing with a large print farm for verification of tank performance from a new manufacturer.
- Provided technical instruction and documentation to support the customer service team's field repair efforts.

Independent Researcher - Soft Electronics

Singapore, SG

Bio-Inspired Soft Robotics Lab - National University of Singapore

May 2019 – August 2019

- Built the first fully 3D printed electrostatic soft loudspeaker, validating its viability by characterizing its frequency range.
- Development of a smart material utilizing dielectric elastomer technology and bilayer wrinkling phenomena that can alter its surface texture as needed.

President - UAV Student Team**University of Alberta Aerial Robotics Group - University of Alberta**

Edmonton, AB
September 2015 – May 2018

- Led a multidisciplinary student group of 30 students at the annual AUVSI Student Unmanned Aerial Systems Competition.
- Designed blended wing airframe and payload drop mechanism.
- Integration of onboard and ground systems (telemetry, imaging, power & steering, wifi data transfer).

Undergraduate Researcher - Oil Refinery Data Analytics & CNC

San Antonio, USA

Flexible Manufacturing and Lean Systems Lab - University of Texas at San Antonio

Summer 2018

- Statistical analysis (Regression analysis, ARIMAX model fitting and forecasting) to predict incidents occurring in refineries.
- Prepared detailed reports detailing predictions of statistical forecasting for oil and gas clients.

Dean's Research Award Researcher

Edmonton, AB

Department of Mechanical Engineering - University of Alberta

January 2018 - April 2018

- Studied an analytical model of human teeth periodontal ligaments to determine its visco-elastic behavior, producing a mathematical model using a system of second-order differential equations approximating its behavior.

Project Engineer Intern

Fort McMurray, AB

Syncrude Canada Limited - Mildred Lake Plant

January 2017 – August 2017

- Handling of Requests for Information (RFI) between contractors and our client company coordinating engineering support to overcome unexpected construction roadblocks.
- Aided the redesign of structural supports for HVAC distributor to account for an unexpected 20-degree roof incline.

Undergraduate Researcher

Aachen, Germany

Institute of Textile Technology - RWTH Aachen University

Summer 2016

- Re-designed spin head of polymer melt-spinning apparatus to combat leakage issue due to insufficient internal vessel pressure.
- Conceptual design of automated cleaning mechanism for in-operation industrial process rollers to eliminate residue build-up, using laser ablation, abrasive cleaning mechanism, and adhesion phenomenon.

Shell Canada Limited - Albion Sands Site

Fort McMurray, AB

Mechanical Engineering Intern

May 2015 - December 2015

- Performed Root Cause Failure Analysis (RCFA) on major feed piping determining failure mechanism to be erosion-corrosion. Pushed for pipe material upgrade from carbon steel to 410 Stainless to prevent failure.
- Identified and removed safety hazards such as dead legs in unused piping with potential for static liquid buildup and bursting when frozen. Alterations made through Management of Change system.
- Designed and implemented automated mailbox organization and reporting software through VBA.

ADDITIONAL PROJECTS**Automated Wheelchair Tire Cleaner - ReYu Paralysis Recovery Center**

Edmonton, AB

Simulations Lead, Systems Lead

Fall 2018

- Organized a team of 6 students designing an automated wheelchair cleaner constrained to a dimension of under 3m by 3m capable of cleaning all powered and manual wheelchairs models up to a loading of 400 kg.
- The cleaner costs \$12,000 to manufacture, featuring safety mechanisms, modularity, and ergonomic controls.

Design of Payload Drop Mechanism for UAV - MECE 409

Edmonton, AB

Project Lead

Fall 2018

- Designed payload drop mechanism housed in modified off the shelf Skywalker X8 airframe featuring 3D printed modular structure and payload bay doors
- Mechanism rated to support up to 2.6 kg. Modified airframe to maintain a static center of gravity before and after drop, ensure stall speed of UAV with bay doors open does not increase by more than 20%, and power consumption of UAV after modification is less than 10%.

Finite Element Analysis of Handled Coffee Sleeve - MECE 563

Edmonton, AB

Project Manager, Design Verification Lead

Fall 2017

- Coordinated team of 4 students designing a cardboard coffee sleeve that features a handle rated for 1-time use
- Conducted a comprehensive finite element analysis using the Tsai-Wu failure criterion concluding the sleeve can support a large coffee to an upward acceleration of 2.25 g-force or 7.5 km/h in running speed
- Performed model verification testing that confirmed FEA simulation model to be 100% correct.

PRESENTATIONS, POSTERS, AND PAPERS

Cheung, J. M., Gao, J., Hinrichsen, B., Ma, D., Nguyen, D., Augustine, R., ... Elliott, D. G. (2018). *University of Alberta Aerial Robotics Group Unmanned Aerial System: 2018 AUVSI SUAS Competition Paper*. Paper and poster presented at the 16th annual Student Unmanned Aerial Systems Competition of the Association for Unmanned Vehicle Systems International Seafarer Chapter, Patuxent River, MD.